Ajay Venkatesh

+1 516 585 9013

Skills

Programming Languages: Python, Java, Scala, R, SQL, C++, TypeScript, JavaScript (ES6), Matlab
Big Data & Processing: PySpark, Apache Spark, Pandas, NumPy, ETL/ELT Workflows, Data Pipelines
Databases: SQL Server, MySQL, PostgreSQL, DynamoDB, MongoDB, Dataverse
Cloud & Warehouses: AWS (Redshift, S3, EC2, Lambda, Glue, CloudWatch, SageMaker, QuickSight, CDK), GCP (BigQuery), Azure (Data Lake, Function Apps)
Tools: Git, Docker, Kubernetes, Jupyter, Postman, Power BI, Visual Studio, Cursor, Operations, Tableau, SAS, Minitab, SAP
AI/ML: TensorFlow, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL.

Experience

Campus Mesh — **Jan 2026 – Present**
Data Engineer — *New York, US*

- Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users** across study groups, messaging, notifications, and profile management, backed by a **MongoDB** store.
- Designed a personalized **recommendation engine** using collaborative filtering on behavioral event data with **Python** feature pipelines, increasing engagement by **28%** and retention by **50%**.
- Engineered **LLM-based retrieval pipelines (RAG)** for CampusIQ, building ingestion/indexing workflows for structured data into an **embedding-based semantic search** store, improving accuracy by **30%**.

Amazon — **May 2025 – Aug 2025**
Software Development Engineer Intern — *Seattle, US*

- Architected a secure data console for Amazon fulfillment center teams to query image/video evidence across FCs; presented **system design** for approval and set up Role Based Access **(4+ roles)** via **OAuth, Midway SSO, HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, **DynamoDB, S3**, IAM, **CloudWatch**, cutting provisioning time **70%** with **multi-stage, multi-region CI/CD**.
- Engineered distributed backend services in **Java** using Coral, Smithy, Dagger, CBOR; applied **low-level design patterns**, cutting execution time **35%** with unit/integration tests via **Mockito**.

New York University (Novel AI Technologies, Inc.) — **Aug 2024 – Present**
Data Engineer — *New York, US*

- Architected an **NSF-funded ($100K)** AI-powered health platform for **iOS** and **Android**, ingesting real-time wearable telemetry into cloud storage for downstream AI model inference on health metrics and food images.
- Engineered **ETL/ELT workflows** with **Python, PySpark, SQL**, transforming **NoSQL** event data into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards.
- Built a full-stack insurance analytics platform deployed via **Docker** containers on **EC2**, and implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue across **6 clients**.
- Implemented **document/media processing pipelines** with **WebSocket real-time data streams** to streamline underwriting via a broker portal integrated with insurer platforms.

PricewaterhouseCoopers (PwC) — **Aug 2021 – Aug 2024**
Data Engineer — *Bengaluru, India*

- Developed a **distributed data processing backend** on **Microsoft Azure** integrating with **Microsoft CRM**, with **event-driven ingestion** and fault-tolerant retries over **SQL Server** stores.
- Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction from unstructured documents into downstream warehouses, reducing processing time by **75%**.
- Optimized batch ingestion with **parallel processing** for **100K-row** Excel feeds; reduced execution from **5 hours to 1 hour**.
- Built **Power BI** dashboards over datasets joined across multiple relational databases, improving reporting efficiency by **40%**.

Projects

Advanced Expense Tracker (Python Data Pipeline Application) | ***Python, SQLite3, Pandas, Matplotlib, Tkinter***

- Architected a modular **Python application** supporting individual and group expense data with secure login, applying **OOP principles** and **MVC & DAO design patterns** over a **SQLite3** persistence layer.
- Implemented concurrent transaction handling and real-time cost aggregation using **Python threading** and locks to ensure thread-safe database writes; visualized spending trends with **Pandas**-backed analytics and Matplotlib charts.

Flight Delay Prediction (Big Data) | ***Hadoop HDFS, PySpark, Spark SQL, Spark MLlib, Seaborn, Matplotlib, Plotly***

- Built a distributed big data pipeline over **Hadoop HDFS** ingesting **2024 BTS US flight data**; used **PySpark** and **Spark SQL** for cleaning, feature engineering, and large-scale aggregations.
- Trained **Spark MLlib** classification models to predict delay severity and surfaced feature importance and delay distributions via **Seaborn, Matplotlib, and Plotly** visualizations.

Education

New York University | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** — Sep 2024 – May 2026

- *Coursework:* Big Data, System Design, Machine Learning, Deep Learning, Data Structures

Certifications

AWS Cloud Practitioner	Azure Data Fundamentals	Microsoft Solution Architect Expert
Power BI Data Analyst Associate	Microsoft Developer Associate	Stanford Machine Learning (Coursera)